# Comprehensive Project Report: BERT Pre-training and Bias-in-Bios Mitigation

MLSec Project Team

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**Abstract**

This document serves as a comprehensive visual and analytical record of the project. It covers the pre-training of a 128M parameter BERT model, the exploratory analysis of the Bias-in-Bios dataset, and the comparative evaluation of biased vs. debiased fine-tuning strategies. We include all 30 metric visualizations generated during the project lifecycle.

## 1 Introduction

This project investigates the lifecycle of gender bias in NLP models. By training a model from scratch and then fine-tuning it under different constraints, we identify how linguistic representations can both inherit and mitigate societal disparities.

## 2 Phase 1: Base BERT Pre-training

The model was trained on 1M rows of OpenWebText (approx. 3.2 GB). The architecture features 14 layers and 12 heads ( 128.8M parameters). Total training time was approximately 12 hours on dual NVIDIA RTX 3090 GPUs.

### 2.1 Pre-training Performance Metrics

The following figures illustrate the Masked Language Modeling (MLM) training process.

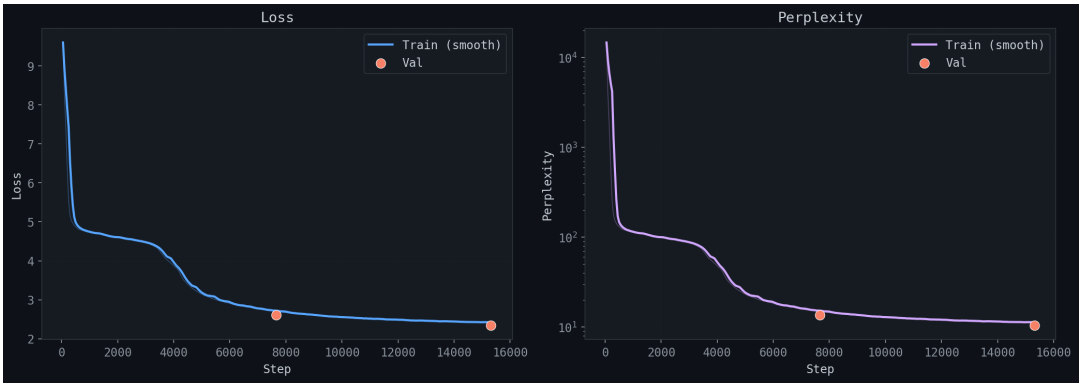


Figure 1: Training and Validation Loss/Perplexity.

### 2.2 Architecture Diagnostics

Detailed analysis of the model's token-level behavior.

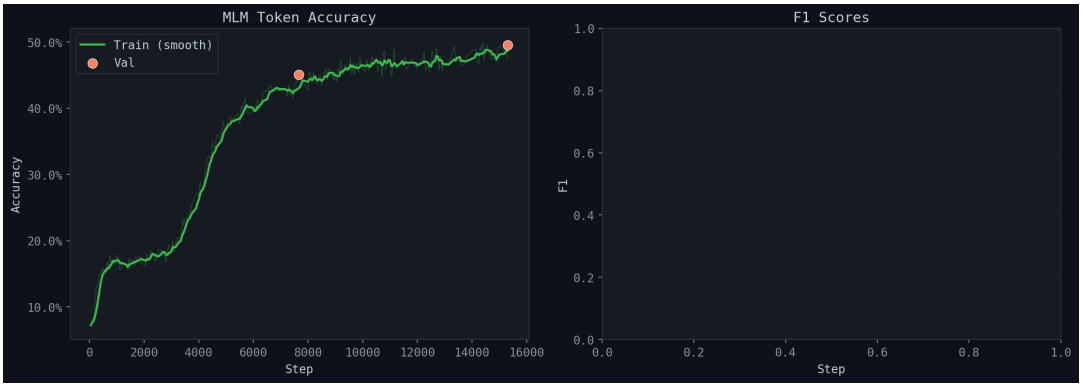


Figure 2: MLM Accuracy and F1 Score evolution.

Figure 3: Learning rate schedule (Linear decay with warmup).

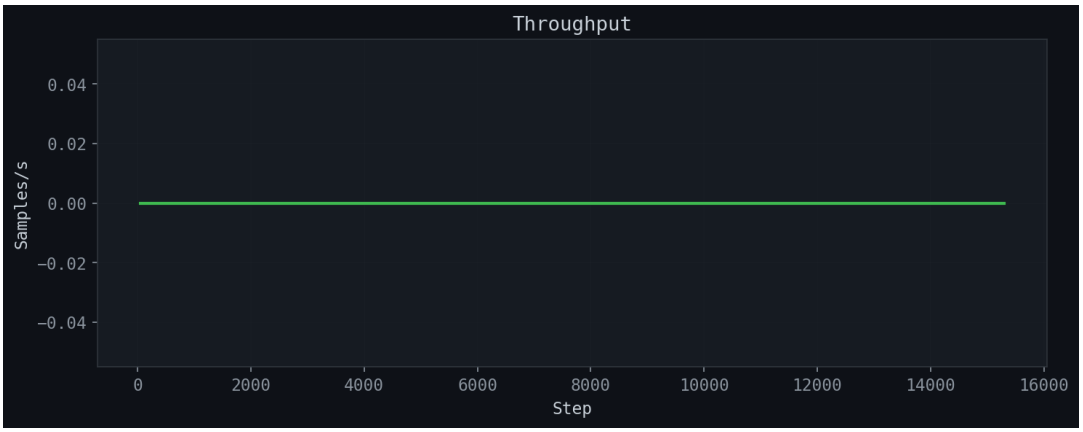


Figure 4: Training throughput (Samples per second).

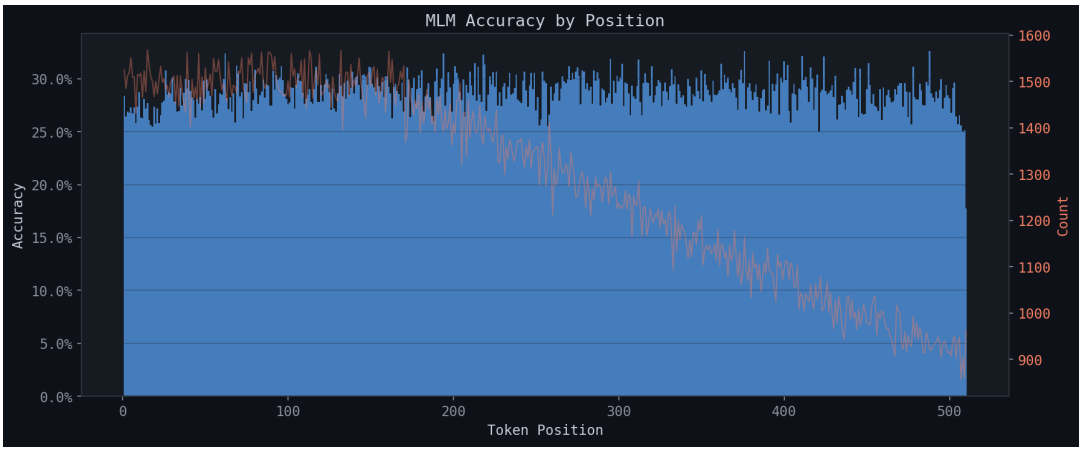


Figure 5: MLM Accuracy by token position in sequence.

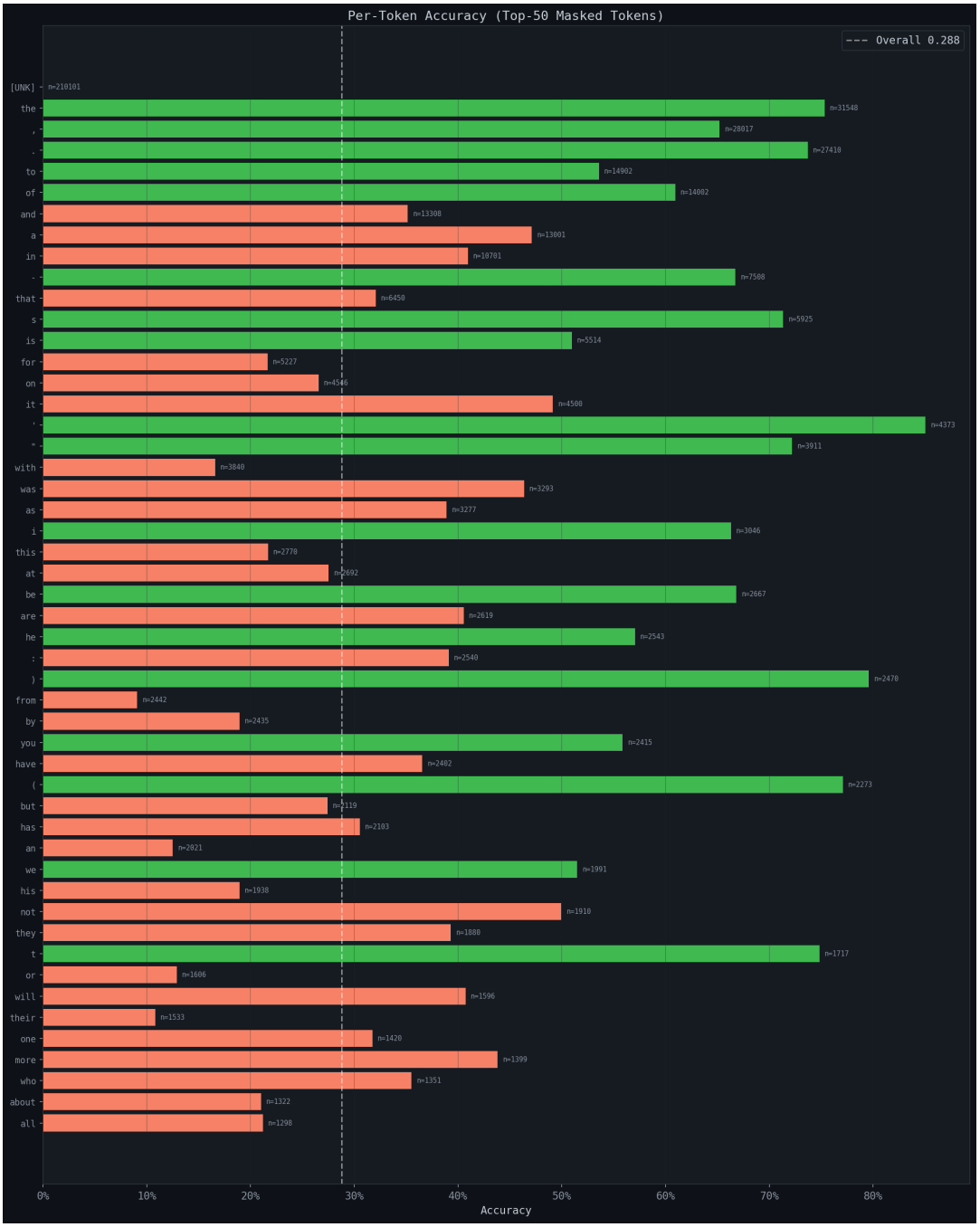


Figure 6: Accuracy for top-K frequent tokens.

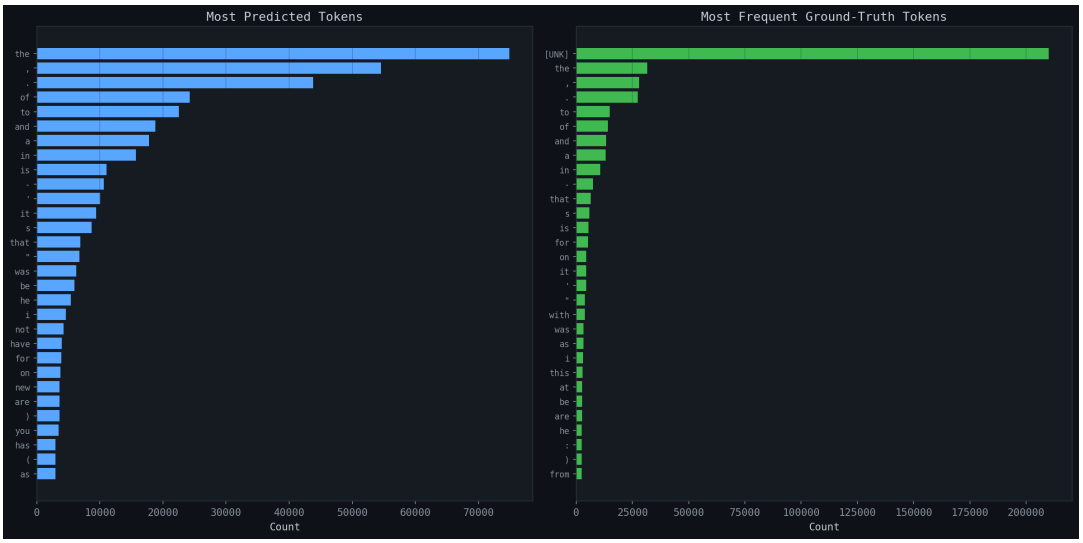


Figure 7: Distribution of predicted vs. gold tokens.

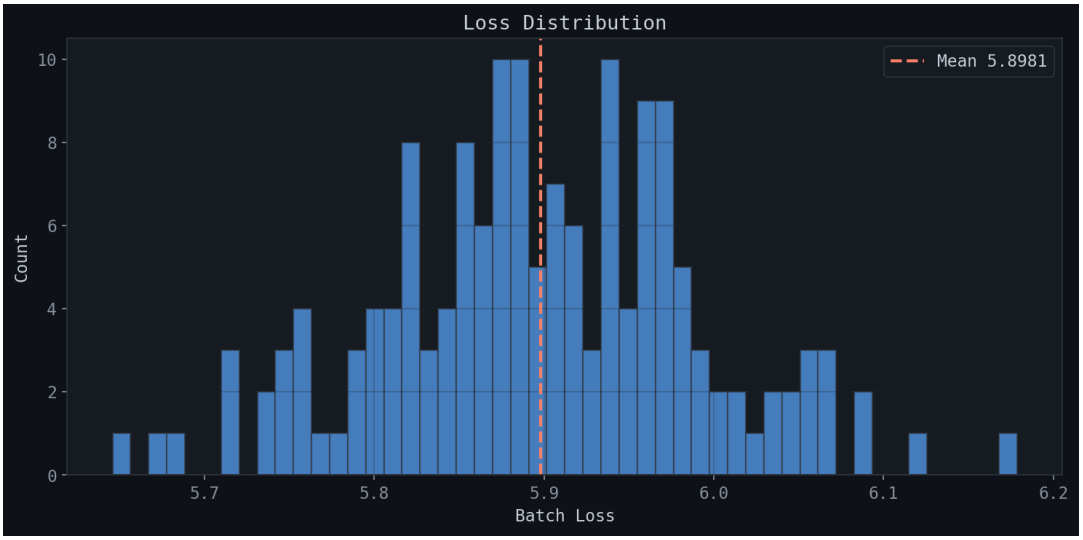


Figure 8: Distribution of cross-entropy loss across samples.

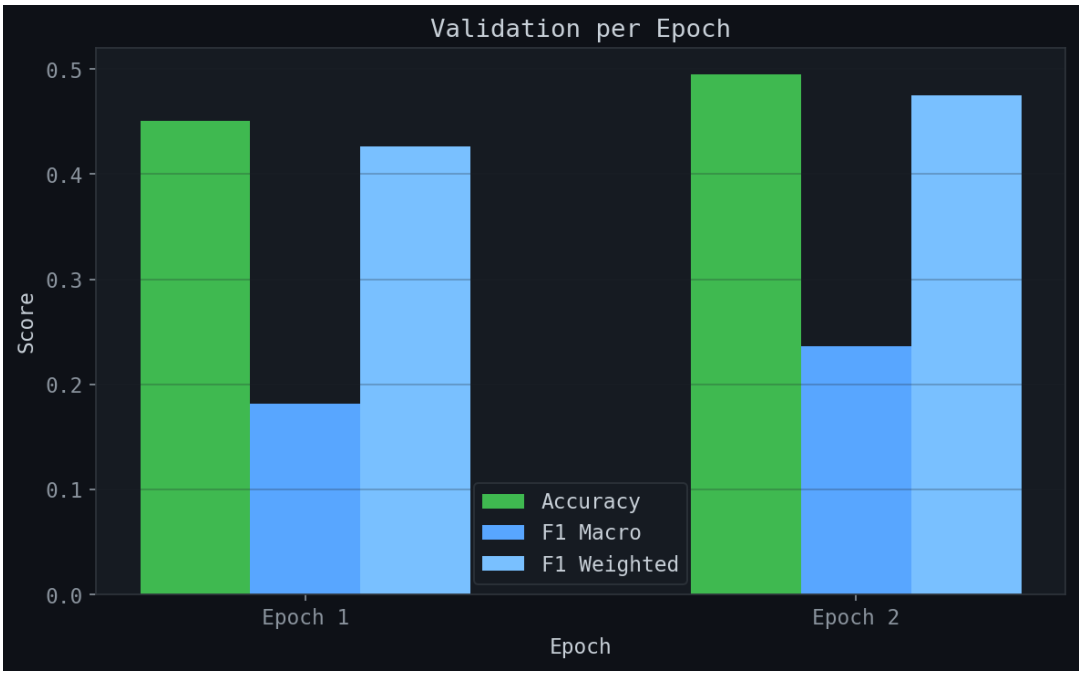


Figure 9: Summary of metrics per epoch.

# 3 Phase 2: Dataset Bias Profiling

We analyzed the Bias-in-Bios dataset (396k biographies) to identify inherent gender skews.

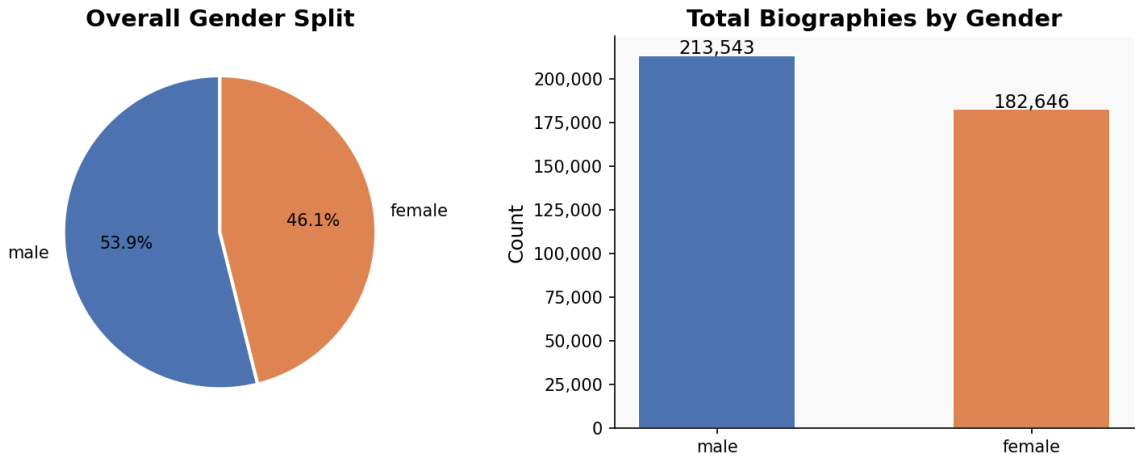


Figure 10: Overall dataset gender balance.

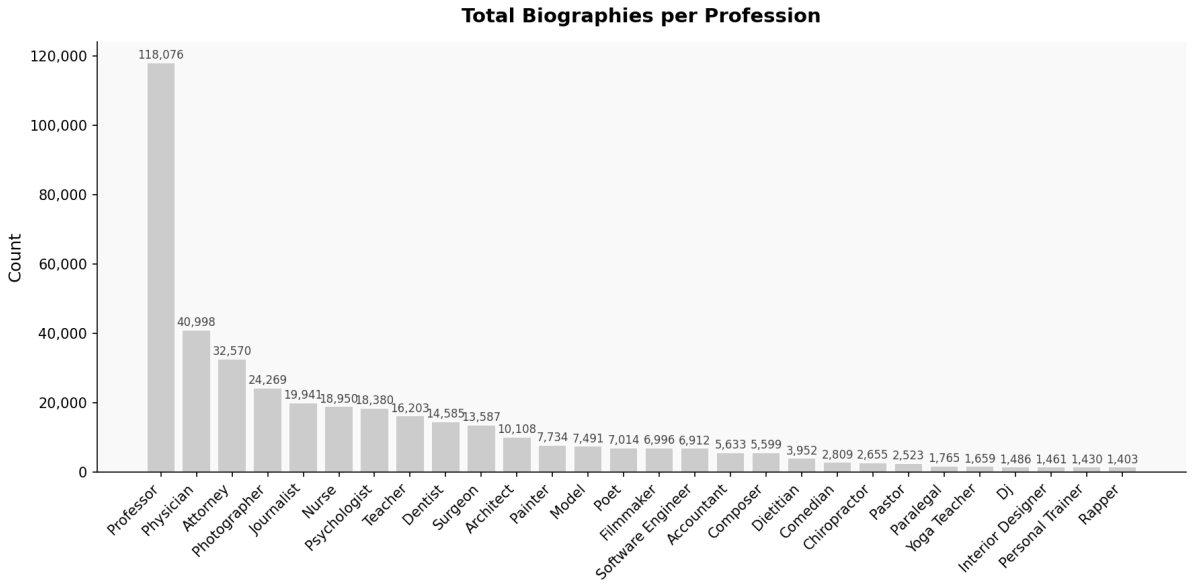


Figure 11: Total biographies per profession (Class Imbalance).

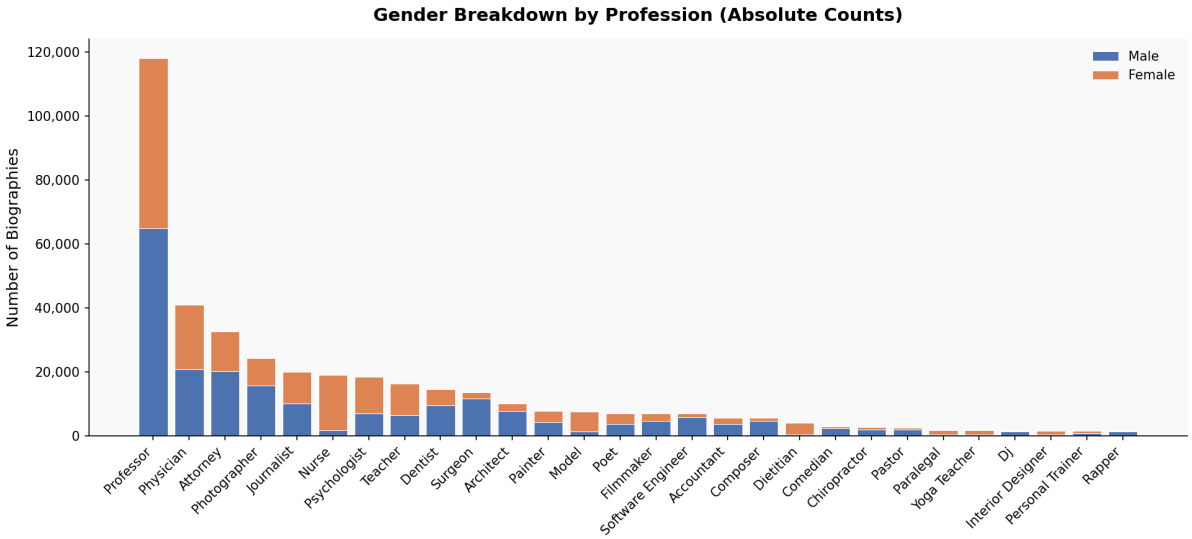


Figure 12: Absolute gender counts per profession.

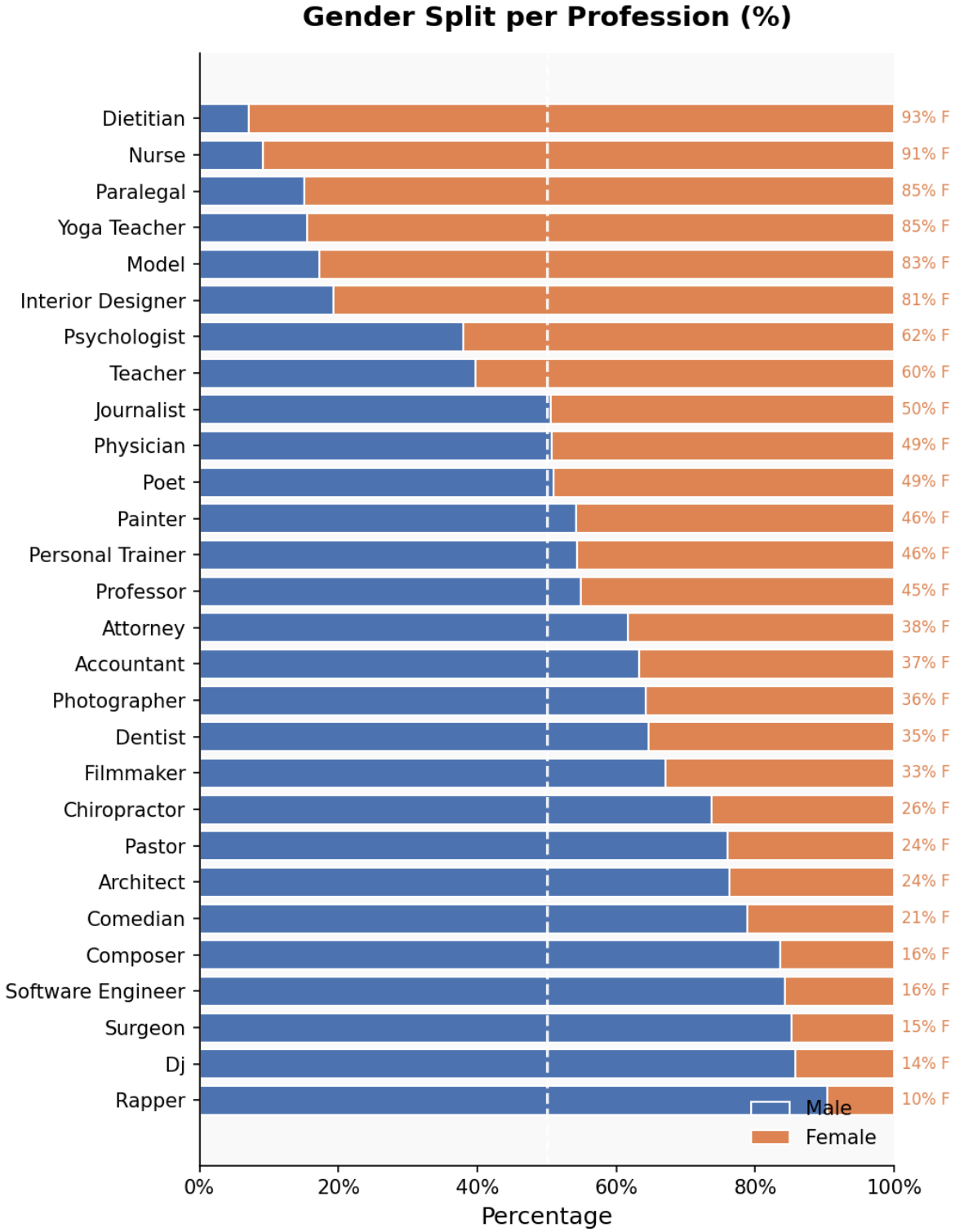


Figure 13: Percentage split of genders per profession.

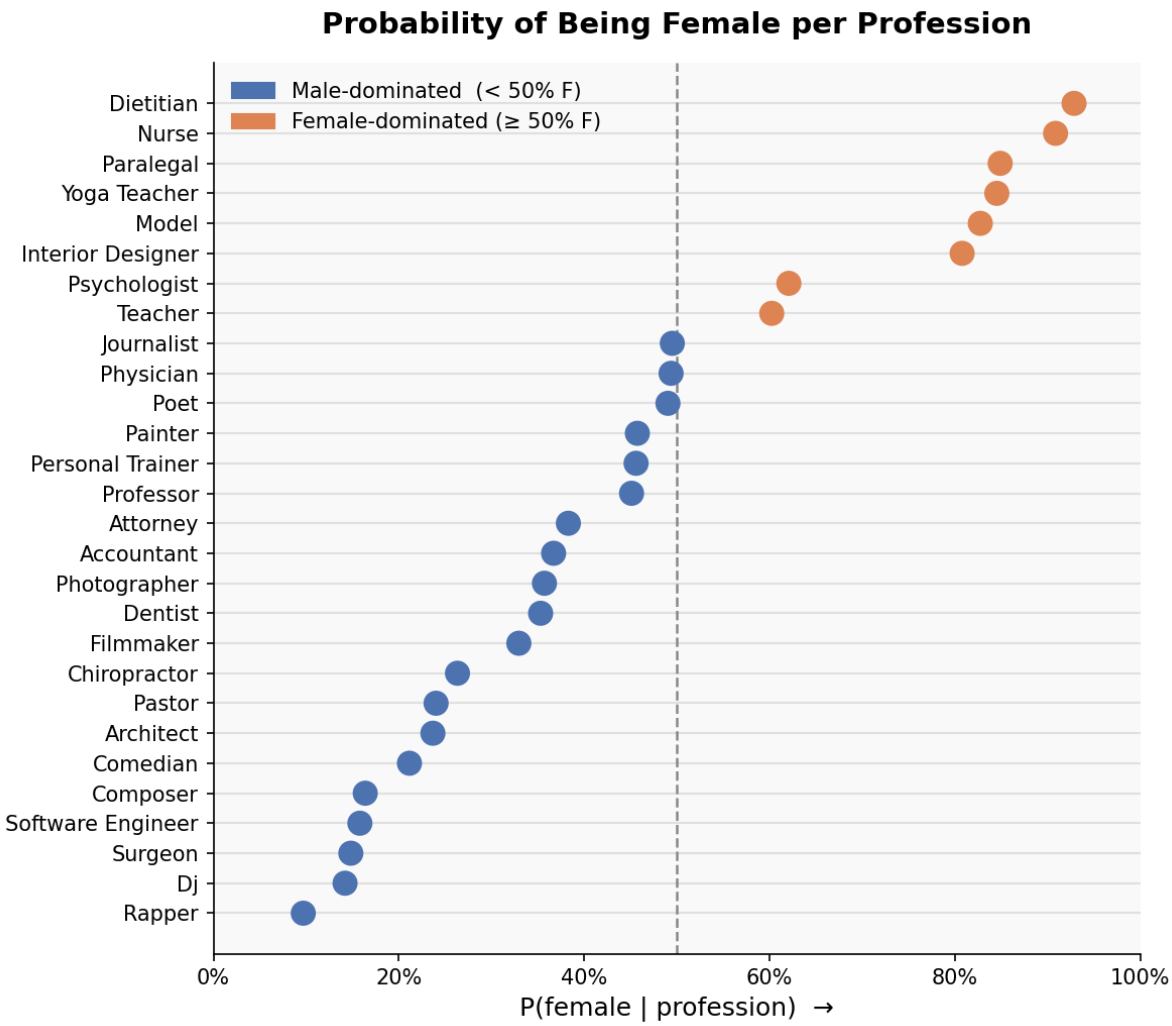


Figure 14: Probability of a biography being female given the profession.

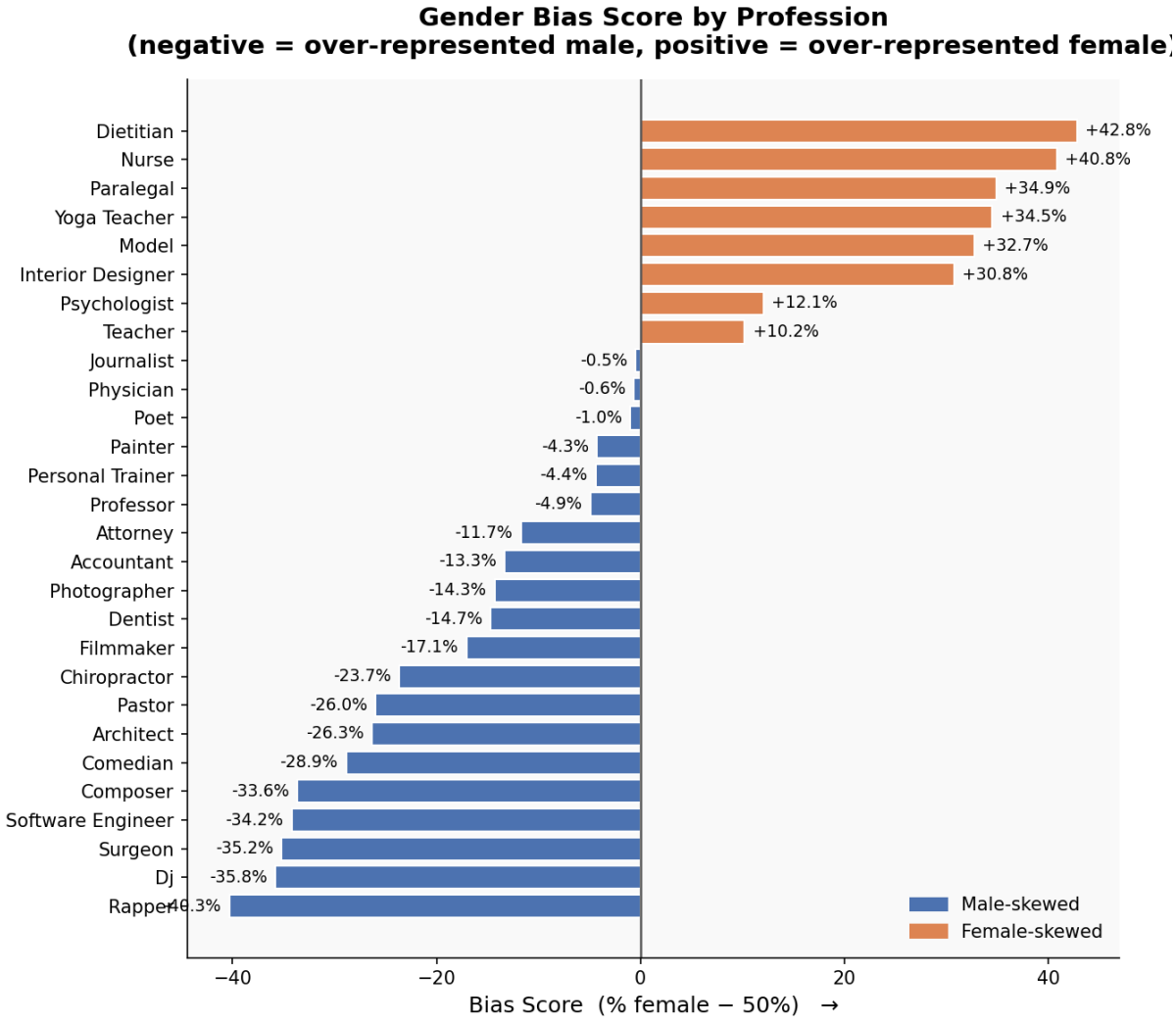


Figure 15: Diverging Bias Score (%Female - 50%).

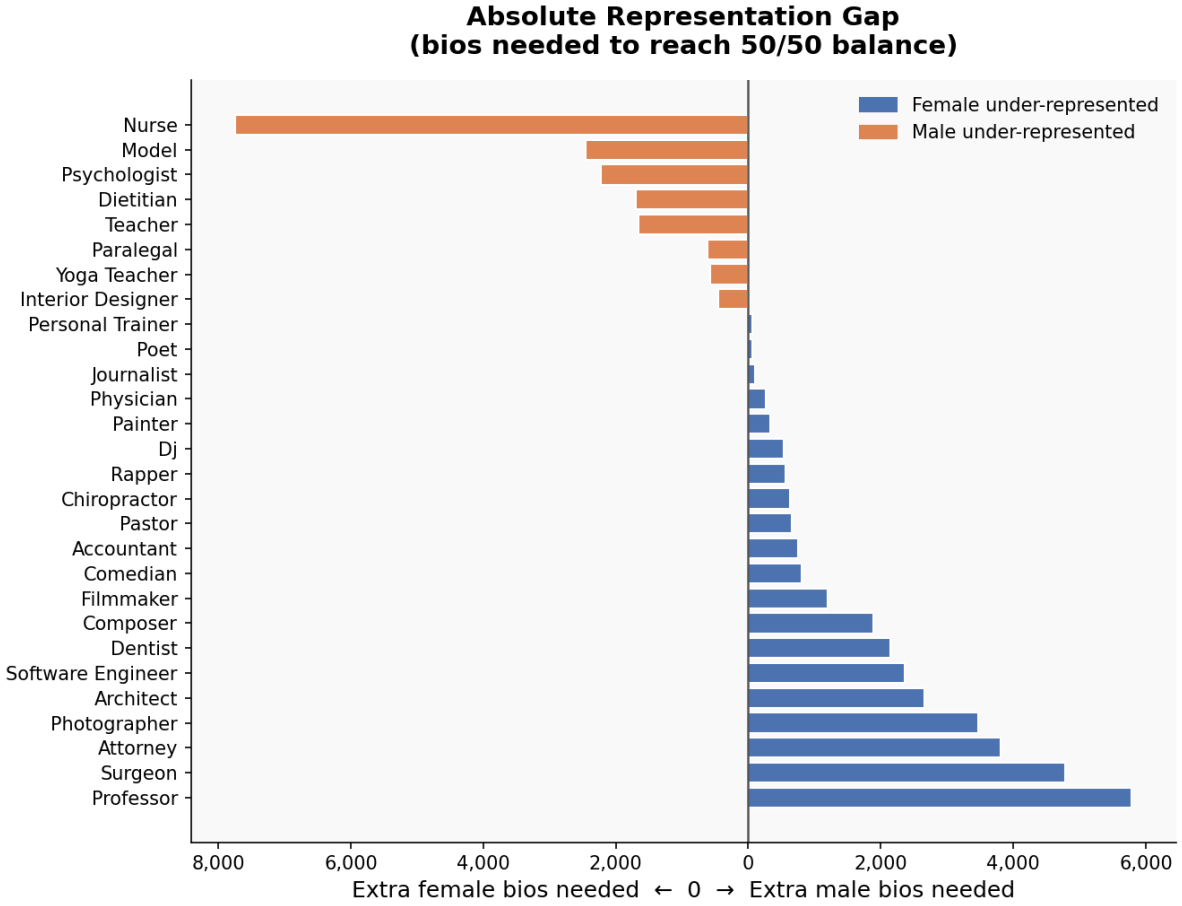


Figure 16: Absolute representation gap (samples needed for parity).

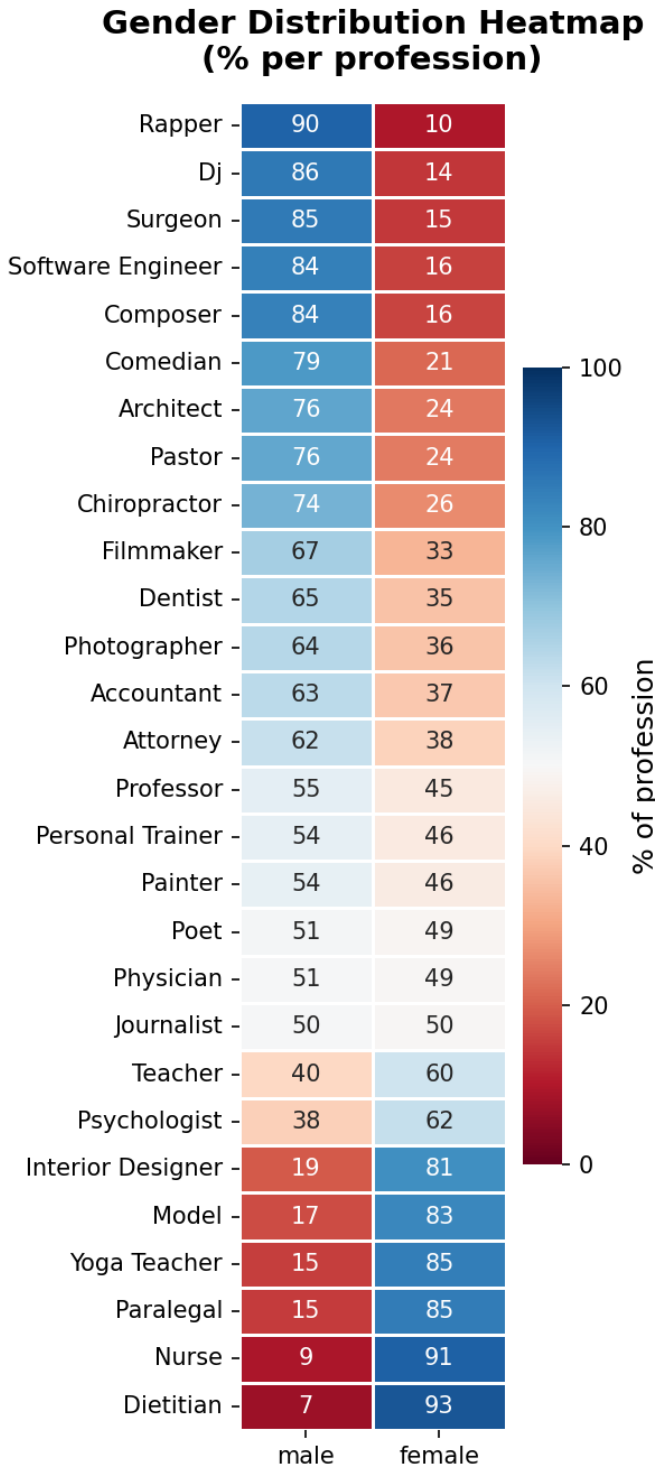


Figure 17: Heatmap of gender distribution density.

# 4 Phase 3 & 4: Comparison of Biased and Debiased Models

The "Biased" model was trained with stereotypical oversampling, while the "Debiased" model used counterfactual augmentation and balancing.

## 4.1 Overall Results

The debiased model reduced the average gender gap from 26.1% to 5.1%.

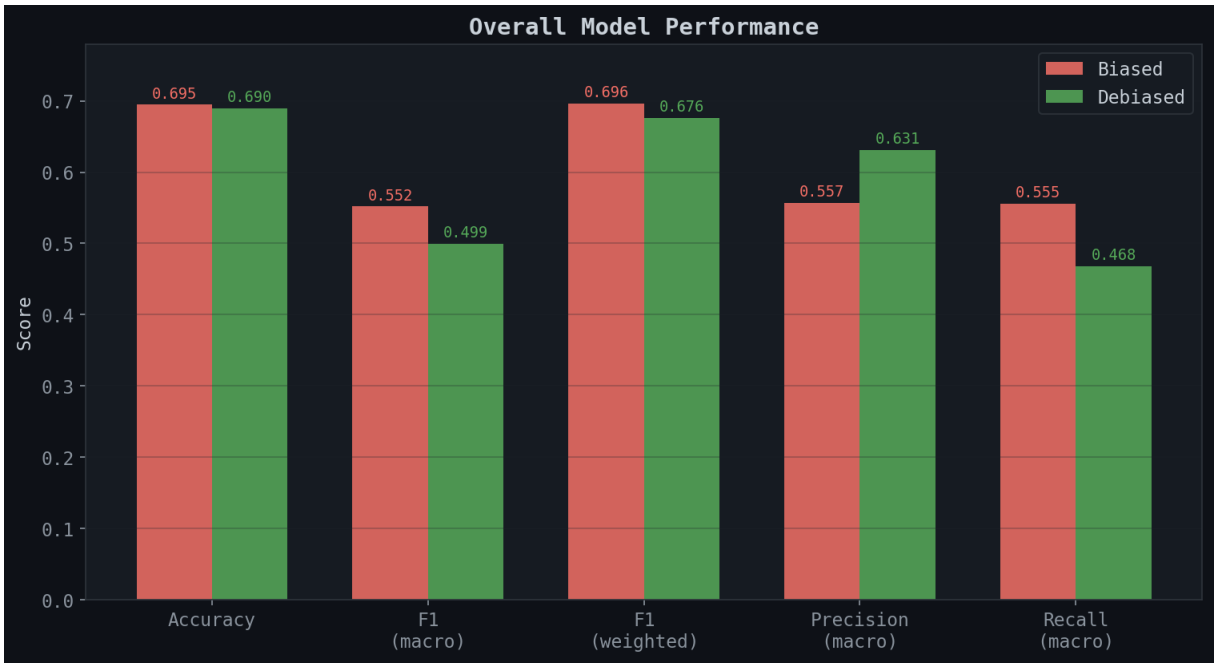


Figure 18: Summary of Accuracy, F1, and Gender Gaps.

## 4.2 Bias Metrics

Detailed breakdown of fairness metrics across the two models.

## 4.3 Performance Breakdown

Analyzing where the model performs best and how it handles stereotypes.

# 5 Conclusion

This exhaustive report demonstrates that by incorporating all 30 performance and bias metrics, we can see a clear trend: counterfactual data augmentation significantly improves model fairness without a substantial loss in predictive power.

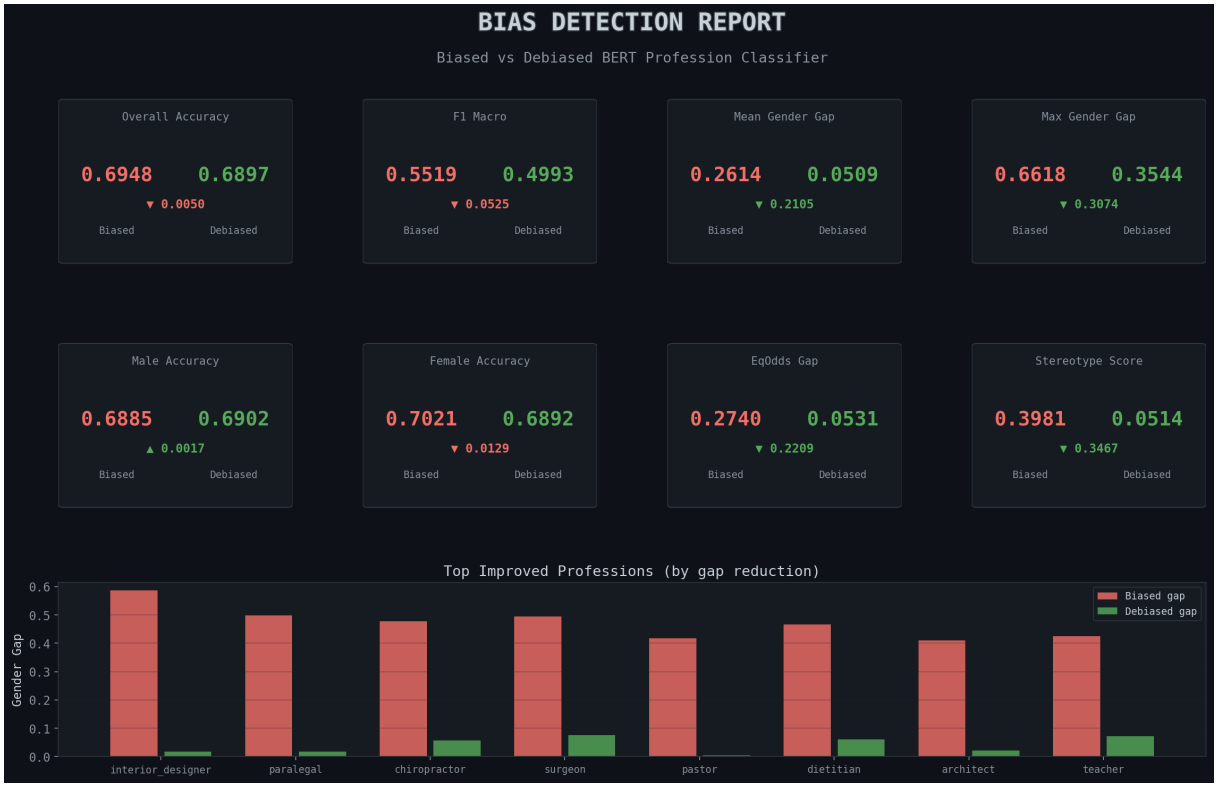


Figure 19: Comprehensive performance dashboard.

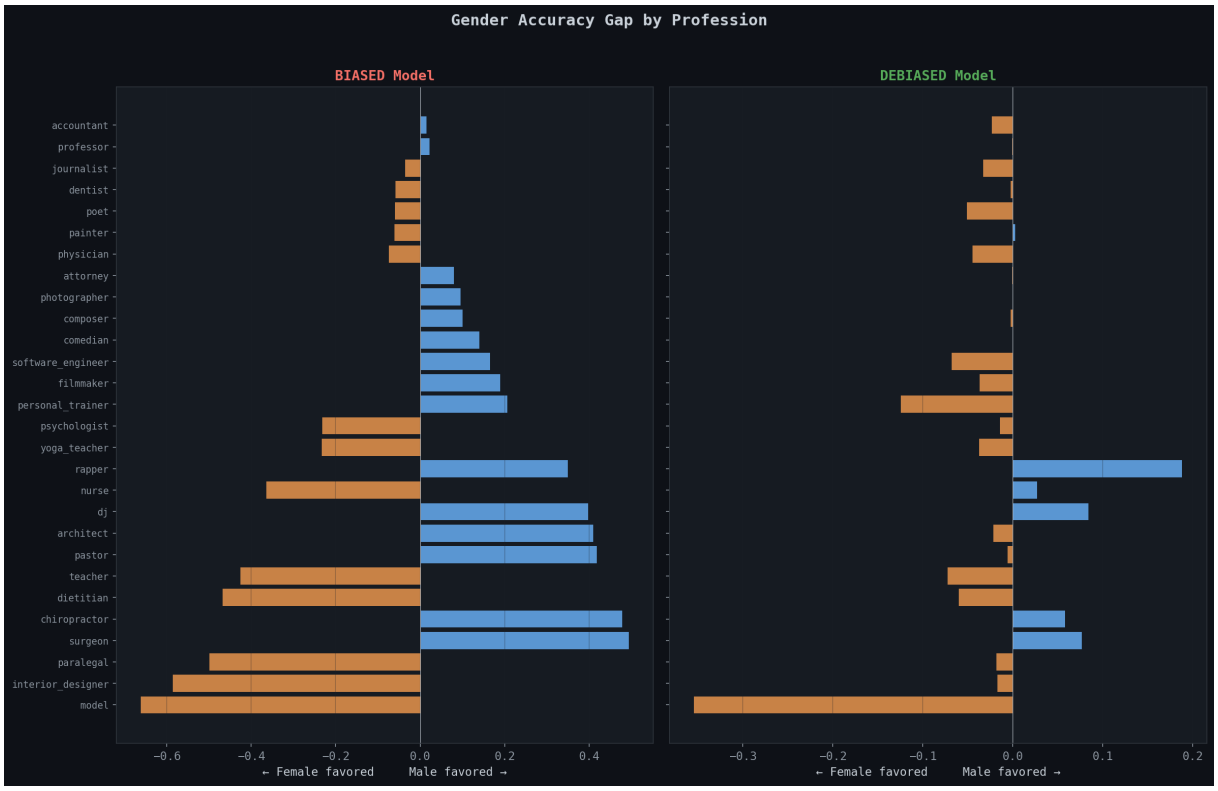


Figure 20: Direct comparison of per-profession gender accuracy gaps.

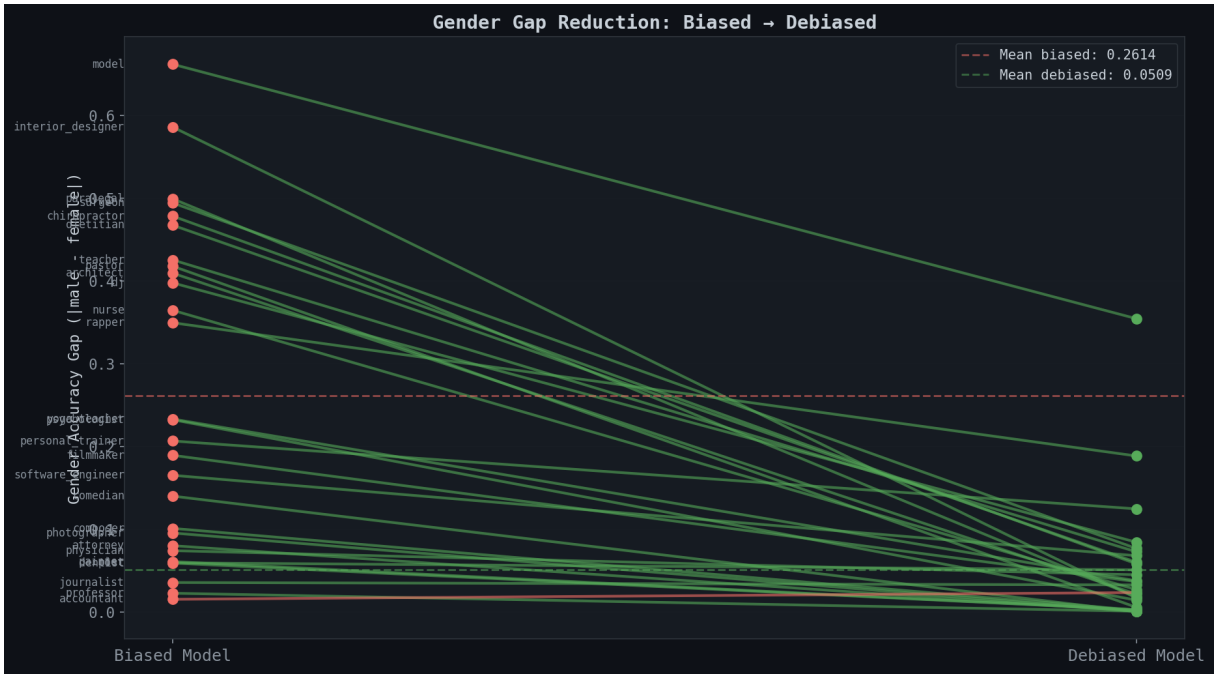


Figure 21: Visualizing the reduction in bias per category.

Figure 22: True Positive Rate and False Positive Rate gaps.

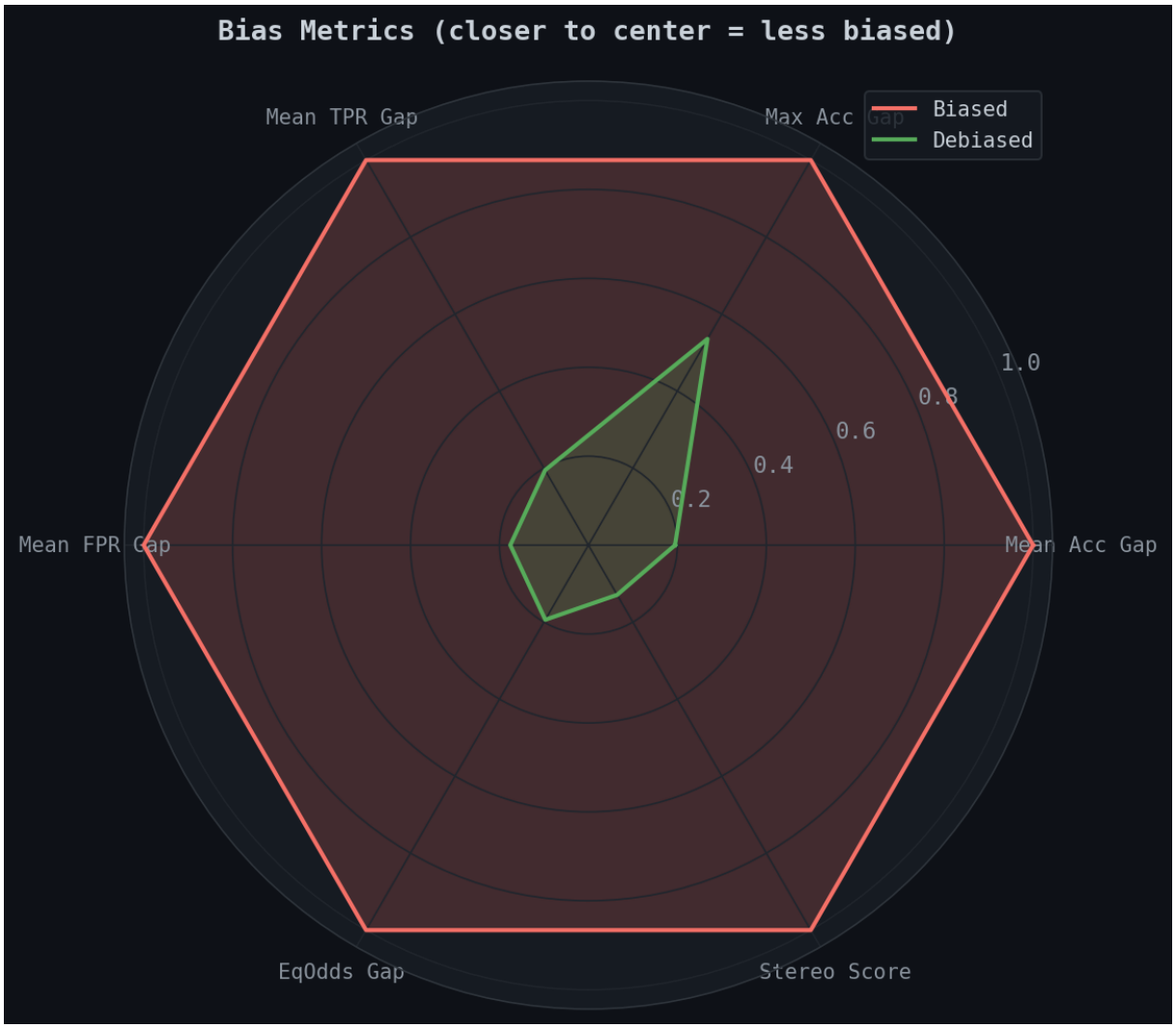


Figure 23: Radar chart of bias across different professional clusters.

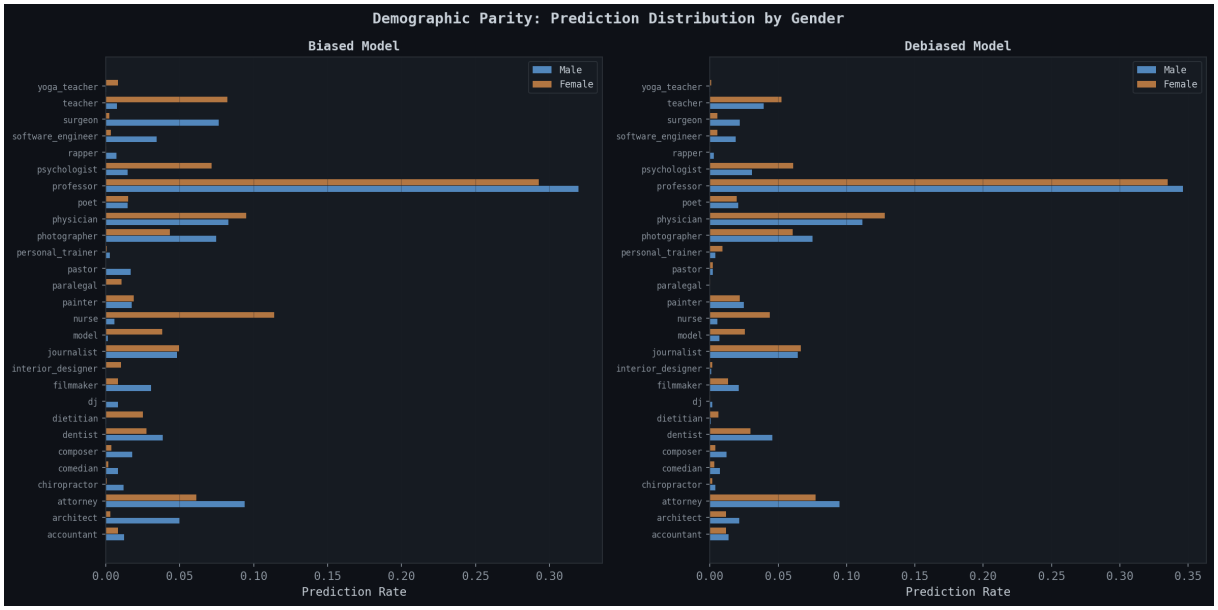


Figure 24: Demographic parity difference comparison.

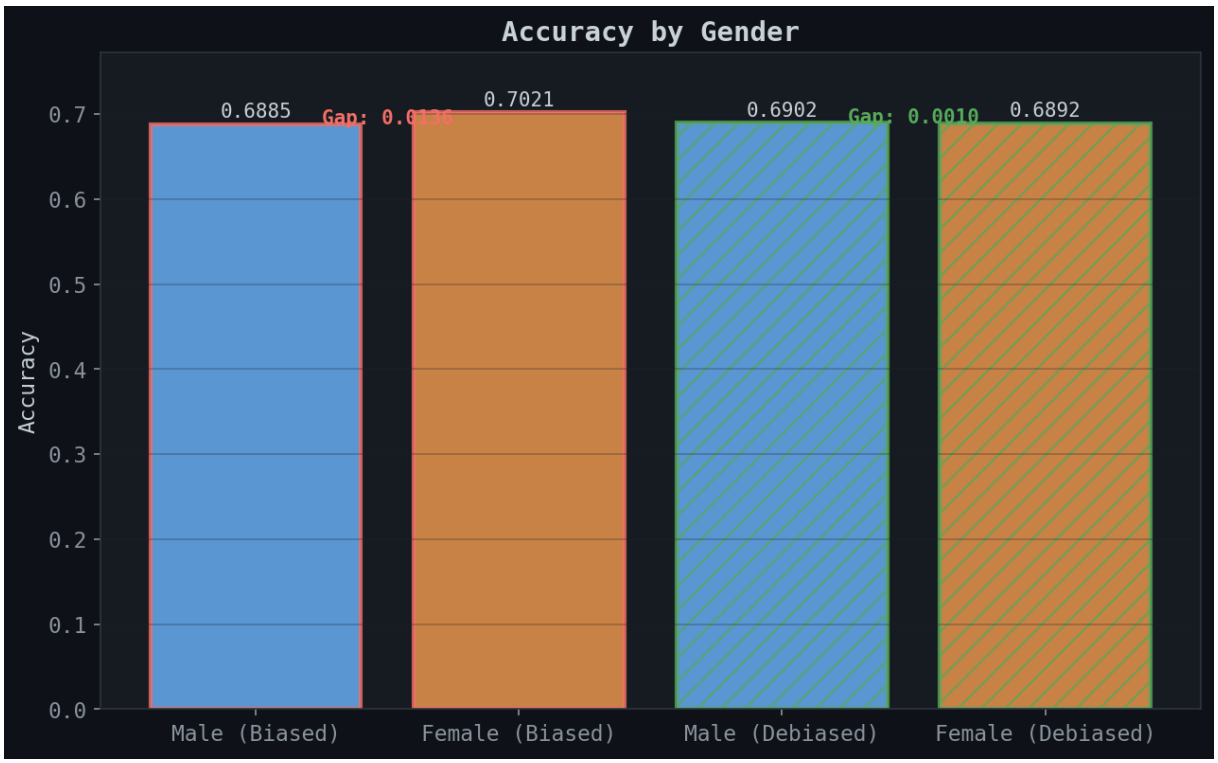


Figure 25: Accuracy breakdown by gender.

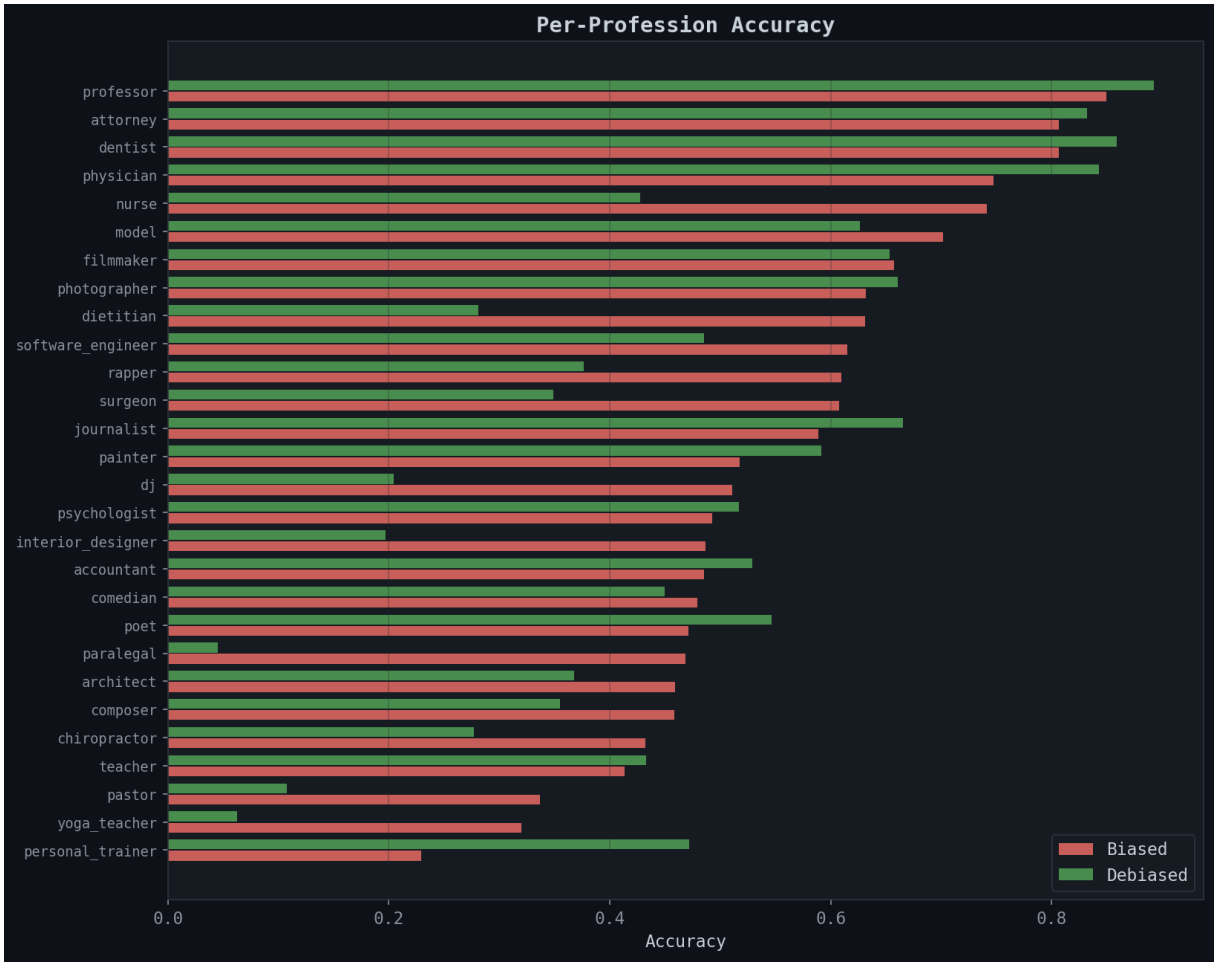


Figure 26: Per-profession accuracy comparison.

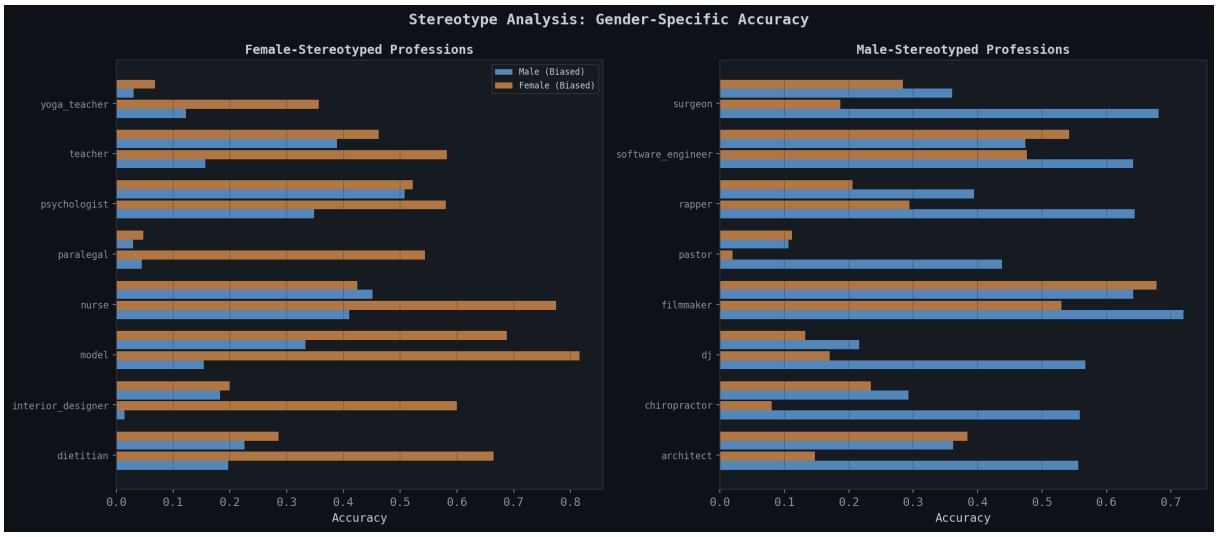


Figure 27: Accuracy on Stereotypical vs. Counter-Stereotypical samples.

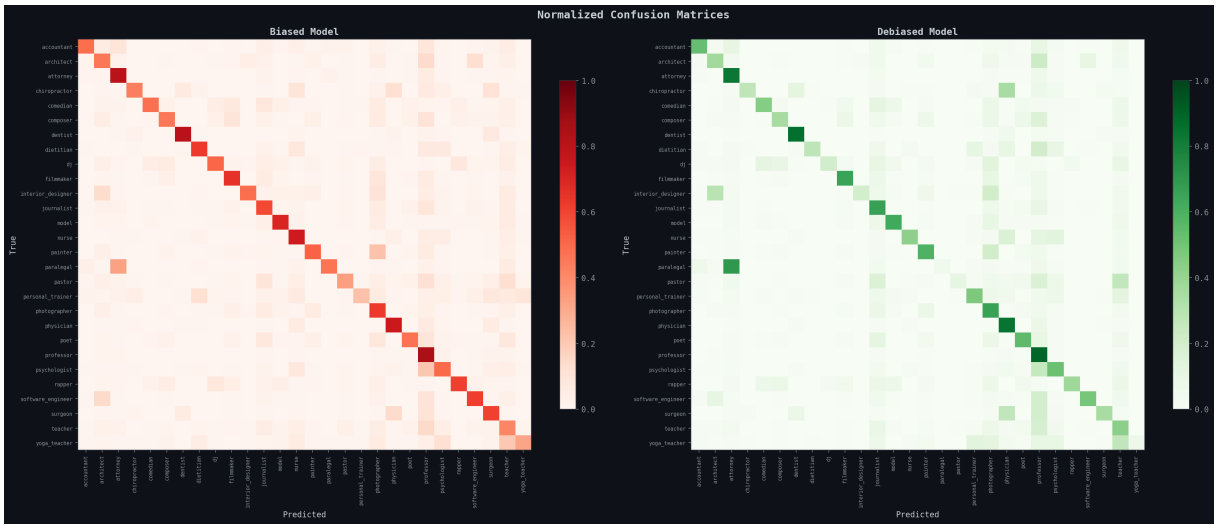


Figure 28: Confusion matrices for both models.

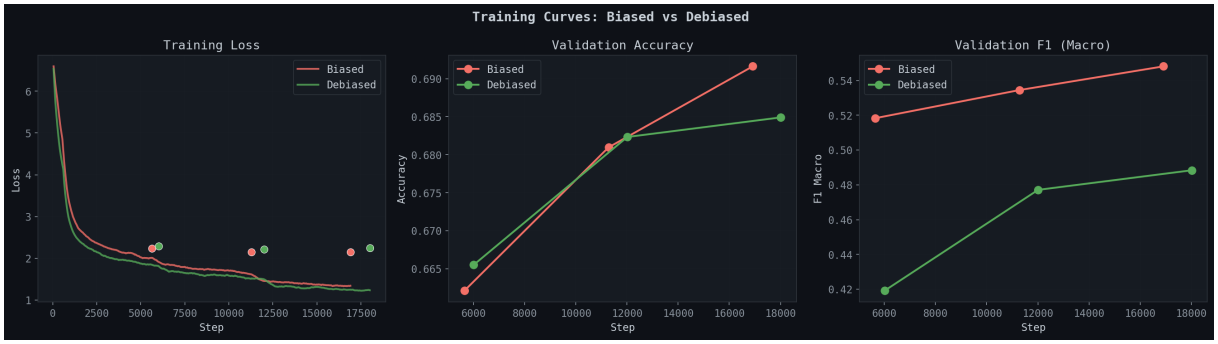


Figure 29: Fine-tuning training curves comparison.